

# Dijkstra Algorithm

## Conditions

- It is important to note the following points regarding Dijkstra Algorithm
- Dijkstra algorithm works only for connected graphs.
- Dijkstra algorithm works only for those graphs that do not contain any negative weight edge.

# Dijkstra Algorithm

- The actual Dijkstra algorithm does not output the shortest paths.
- It only provides the value or cost of the shortest paths.
- By making minor modifications in the actual algorithm, the shortest paths can be easily obtained.
- Dijkstra algorithm works for directed as well as undirected graphs.

# Dijkstra's Algorithm

## Dijkstra's Algorithm

**Step 1.** Set  $\lambda(s) = 0$  and for all vertices  $v \neq s$ ,  $\lambda(v) = \infty$ . Set  $T = V$ , the vertex set of  $G$ . (We will think of  $T$  as the set of vertices “uncoloured”.)

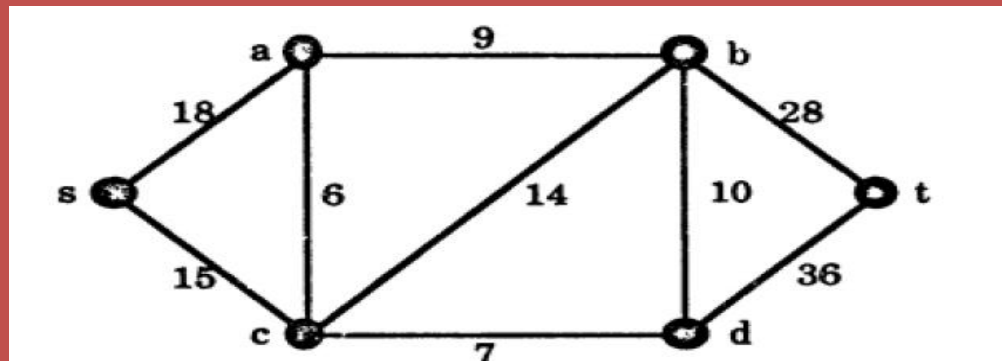
**Step 2.** Let  $u$  be a vertex in  $T$  for which  $\lambda(u)$  is minimum.

**Step 3.** If  $u = t$ , stop.

**Step 4.** For every edge  $e = uv$  incident with  $u$ , if  $v \in T$  and  $\lambda(v) > \lambda(u) + w(e)$  change the value of  $\lambda(v)$  to  $\lambda(u) + w(e)$  (i.e., given an edge  $e$  from an “uncoloured” vertex  $v$  to  $u$ , change  $\lambda(v)$  to  $\min(\lambda(v), \lambda(u) + w(e))$ ).

**Step 5.** Change  $T$  to  $T - \{u\}$  and go to Step 2, (i.e., “colour”  $u$  and then go back to Step 2 to find an “uncoloured” vertex with minimum label).

Q.Using Dijkstra's algorithm find the shortest path and length of the shortest path from the vertex s to the vertex t of the following weighted graph.



# Solution:

Step 1. The initial labelling is given by:

| vertex $v$   | $s$     | $a$      | $b$      | $c$      | $d$      | $t$      |
|--------------|---------|----------|----------|----------|----------|----------|
| $\lambda(v)$ | 0       | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $T$          | { $s$ , | $a$ ,    | $b$ ,    | $c$ ,    | $d$ ,    | $t$ }    |

Step 2.  $u = s$  has  $\lambda(u)$  a minimum (with value 0).

**Step 4.** There are two edges incident with  $s$ , namely  $\{s, a\}$  and  $\{s, c\}$ . Both  $a$  and  $c$  are in  $T$ .

$$\lambda(a) = \min(\lambda(a), \lambda(s) + w(\{s, a\})) = \min(\infty, 0 + 18) = \min(\infty, 18) = 18.$$

$$\lambda(c) = \min(\lambda(c), \lambda(s) + w(\{s, c\})) = \min(\infty, 0 + 15) = \min(\infty, 15) = 15.$$

**Step 5.** Reset  $T = T - \{s\} = \{a, b, c, d, t\}$ . Thus, we have

| vertex $v$   | $s$ | $a$  | $b$      | $c$  | $d$      | $t$      |
|--------------|-----|------|----------|------|----------|----------|
| $\lambda(v)$ | 0   | 18   | $\infty$ | 15   | $\infty$ | $\infty$ |
| $T$          | {   | $a,$ | $b,$     | $c,$ | $d,$     | $t\}$    |

**Step 2.**  $u = c$  has  $\lambda(u)$  a minimum and  $u$  in  $T$  (with  $\lambda(u) = 15$ ).

**Step 4.** There are three edges incident with  $c$ , namely  $\{c, a\}$ ,  $\{c, b\}$ , and  $\{c, d\}$ . Each of  $a$ ,  $b$ , and  $d$  is in  $T$ .

$$\lambda(a) = \min(\lambda(a), \lambda(c) + w(\{c, a\})) = \min(18, 15 + 6) = \min(18, 21) = 18.$$

$$\lambda(b) = \min(\lambda(b), \lambda(c) + w(\{c, b\})) = \min(\infty, 15 + 14) = \min(\infty, 29) = 29.$$

$$\lambda(d) = \min(\lambda(d), \lambda(c) + w(\{c, d\})) = \min(\infty, 15 + 7) = \min(\infty, 22) = 22.$$

**Step 5.** Reset  $T = T - \{c\} = \{a, b, d, t\}$ . Thus, we have

| vertex $v$   | $s$ | $a$  | $b$  | $c$ | $d$  | $t$      |
|--------------|-----|------|------|-----|------|----------|
| $\lambda(v)$ | 0   | 18   | 29   | 15  | 22   | $\infty$ |
| $T$          | {   | $a,$ | $b,$ |     | $d,$ | $t\}$    |

**Step 2.**  $u = a$  has  $\lambda(u)$  a minimum for  $u$  in  $T$  (with  $\lambda(u) = 18$ ).

**Step 4.** There is one edge incident with  $a$ , namely  $\{a, b\}$  with  $b$  is in  $T$ .

$$\lambda(b) = \min(\lambda(b), \lambda(a) + w(\{a, b\})) = \min(29, 18 + 9) = \min(29, 27) = 27.$$

**Step 5.** Reset  $T = T - \{a\} = \{b, d, t\}$ . Thus, we have

| vertex $v$   | $s$ | $a$ | $b$  | $c$ | $d$  | $t$      |
|--------------|-----|-----|------|-----|------|----------|
| $\lambda(v)$ | 0   | 18  | 27   | 15  | 22   | $\infty$ |
| $T$          |     |     | $\{$ |     | $d,$ | $t\}$    |

**Step 2.**  $u = d$  has  $\lambda(u)$  minimum for  $u$  in  $T$  (with  $\lambda(u) = 22$ ).

**Step 4.** There are two edges incident with  $d$ , namely  $\{d, b\}$  and  $\{d, t\}$  with  $b, t$  are in  $T$ .

$$\lambda(b) = \min(\lambda(b), \lambda(d) + w(\{d, b\})) = \min(27, 22 + 10) = \min(27, 32) = 27.$$

$$\lambda(t) = \min(\lambda(t), \lambda(d) + w(\{d, t\})) = \min(\infty, 22 + 36) = \min(\infty, 58) = 58.$$

**Step 5.** Reset  $T = T - \{d\} = \{b, t\}$ . Thus, we have

| vertex $v$   | $s$ | $a$ | $b$ | $c$ | $d$ | $t$   |
|--------------|-----|-----|-----|-----|-----|-------|
| $\lambda(v)$ | 0   | 18  | 27  | 15  | 22  | 58    |
| $T$          | {   |     |     |     |     | $t\}$ |

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**Step 2.**  $u = b$  has  $\lambda(u)$  minimum for  $u$  in  $T$  (with  $\lambda(u) = 27$ ).

**Step 4.** There is one edge incident with  $b$ , namely  $\{b, t\}$  with  $t$  in  $T$ .

$$\lambda(t) = \min(\lambda(t), \lambda(b) + w(\{b, t\})) = \min(58, 27 + 28) = \min(58, 55) = 55.$$

**Step 5.** Reset  $T = T - \{b\} = \{t\}$ . Thus, we have

| vertex $v$   | $s$ | $a$ | $b$ | $c$ | $d$ | $t$   |
|--------------|-----|-----|-----|-----|-----|-------|
| $\lambda(v)$ | 0   | 18  | 27  | 15  | 22  | 55    |
| $T$          | {   |     |     |     |     | $t\}$ |

Step 2.  $u = t$ , the only choice.

Step 3. Stop.

When the algorithm stops the values  $\lambda(v)$  give the lengths of the shortest paths from the vertex  $s$  to each vertex  $v$ . Thus the lengths of these paths from  $s$  to  $a, b, c, d, t$  are 18, 27, 15, 22 and 55 respectively.